Ion and photon pulses per readout cell

NAUSICAA generated report

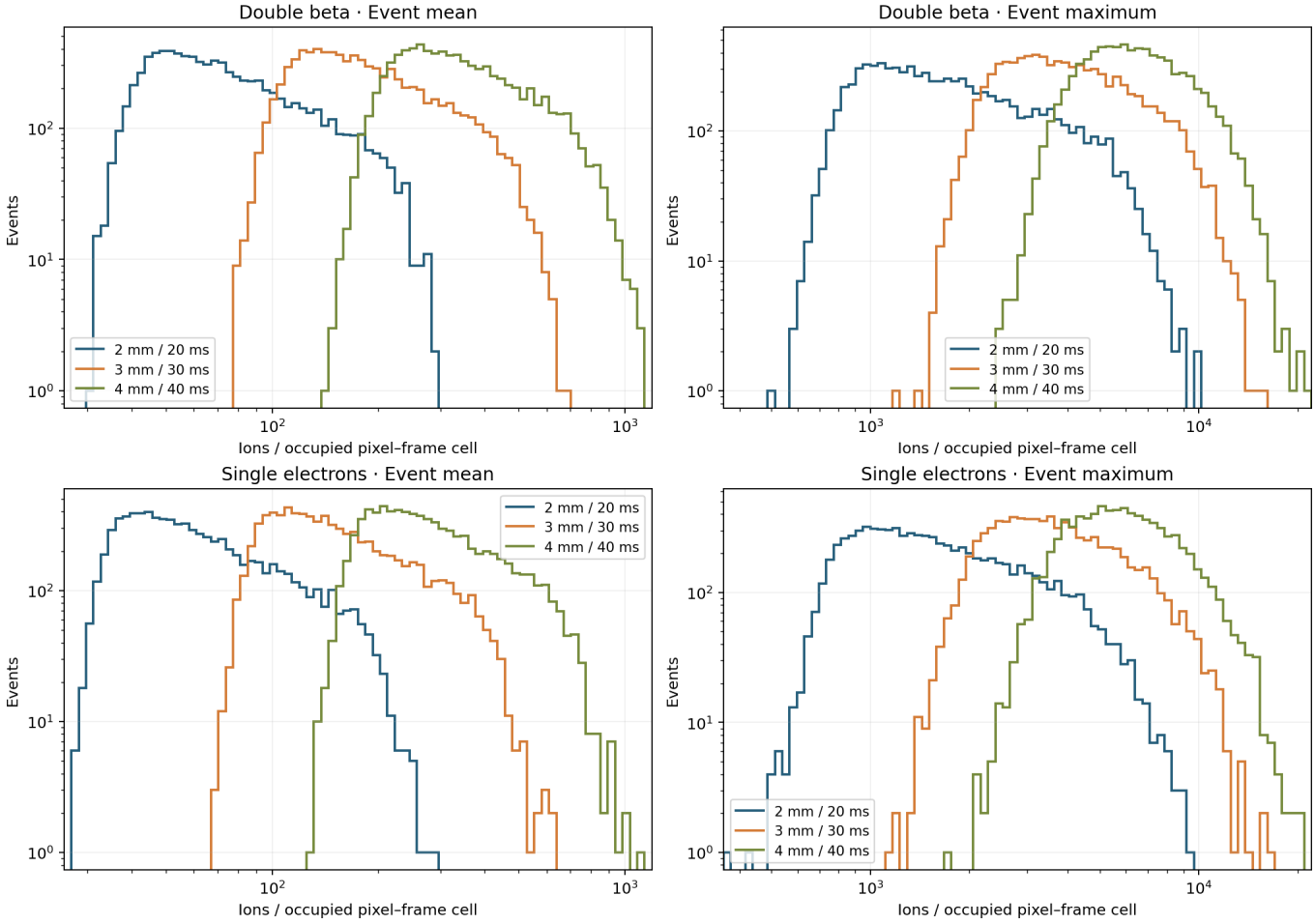
Sample and assumptions

The 1 mm Geant4 source samples contribute 8,000 events each. The same Fano-sampled primary pairs and event placement feed all branches. W = 21.90 eV, F = 0.150; electron lifetime = 15.0 ms, E/P = 3.0 kV/cm/bar, J = 0.010. All occupied cells and all events with pulses enter the distributions; no count cut is applied. Carriers were sampled directly into physical readout cells, with no diffusion-histogram truncation. No time-integrated pixel or SiPM charge enters these plots. The ion and photon ledgers are exact against their saved histories.

Ion pixel–frame pulses

An ion pulse is one pixel–frame cell with positive ion count. The three ion grids use 2, 3 and 4 mm transverse pitch and 20, 30 and 40 ms frames, respectively, corresponding to 2, 3 and 4 mm along drift. Every event mean is its total ions divided by its number of occupied pixel–frame cells; the event maximum is the largest cell count.

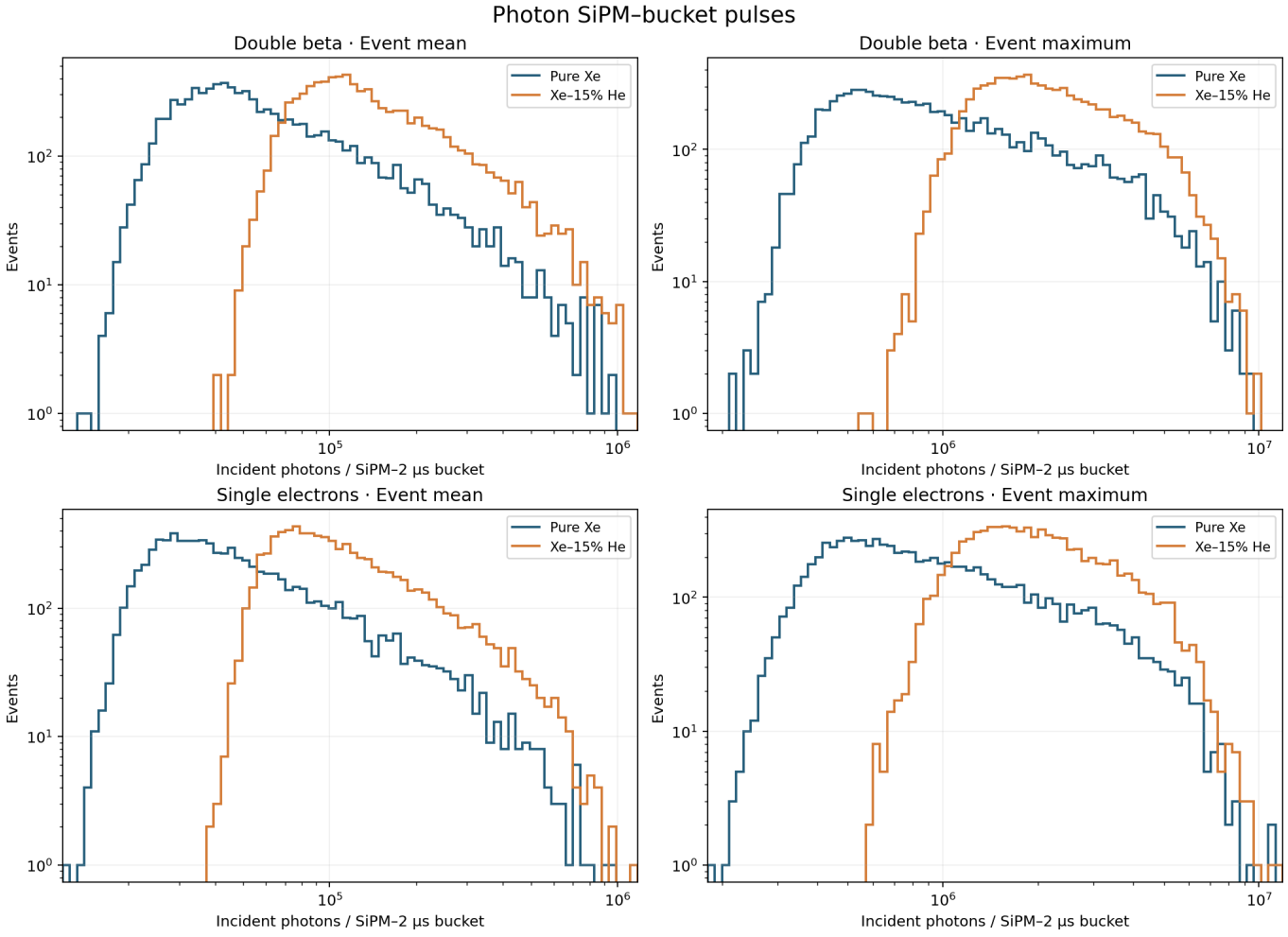
Species	Setting	Events	Occupied cells	Total signal	Mean of means	Median mean	Mean maximum	Median maximum
Double beta	2 mm / 20 ms	8,000	13,474,452	897,306,359	81.89	67.21	2,027.4	1,570.0
Double beta	3 mm / 30 ms	8,000	5,326,135	897,306,359	200.25	170.58	4,342.7	3,787.0
Double beta	4 mm / 40 ms	8,000	2,828,841	897,306,359	367.58	321.51	6,759.6	6,245.0
Single electrons	2 mm / 20 ms	8,000	16,024,040	897,244,198	69.01	55.72	1,881.9	1,461.0
Single electrons	3 mm / 30 ms	8,000	6,386,831	897,244,198	167.14	140.46	4,007.4	3,475.5
Single electrons	4 mm / 40 ms	8,000	3,412,957	897,244,198	304.33	264.11	6,236.0	5,722.0


Ion pixel–frame pulses
Double beta · Event mean
Double beta · Event maximum
Single electrons · Event mean
Single electrons · Event maximum
Events
Ions / occupied pixel–frame cell
2 mm / 20 ms
3 mm / 30 ms
4 mm / 40 ms

Photon SiPM–bucket pulses

A photon pulse is one SiPM–2 μs bucket with positive incident VUV photon count. Pure Xe and Xe–15% He use the same 10 mm hole pitch and electron settings. Every event mean divides its total incident photons by its occupied SiPM–bucket count; the event maximum is the largest bucket count. These are incident photons before SiPM PDE and electronics.

Species	Setting	Events	Occupied cells	Total signal	Mean of means	Median mean	Mean maximum	Median maximum
Double beta	Pure Xe	8,000	7,709,774	381,687,642,280	79,472.22	50,385.34	1,302,701.2	844,057.0
Double beta	Xe–15% He	8,000	3,048,897	375,319,166,815	165,930.15	123,124.19	2,453,245.4	2,038,311.0
Single electrons	Pure Xe	8,000	9,368,261	381,739,047,046	65,305.89	41,346.55	1,219,160.9	789,159.0
Single electrons	Xe–15% He	8,000	3,731,089	375,368,751,796	135,647.88	100,328.55	2,269,206.3	1,891,749.0



Provenance

Numerical reduction: `readout_pulses.h5`. Run ID: `238ddc5a75e0fd9b`. The HDF5 file preserves per-event values, histogram bins, resolved cards, source/code hashes and charge/light ledgers.